

Antonin Blot

CONTACT INFORMATION 23 Nassington Road
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NATIONALITY French

WORK EXPERIENCE **Senior laboratory research scientist in the lab of Petr Znamenskiy**, The Francis Crick Institute, London

Neural circuit for depth perception in the primary visual cortex **September 2020–present**

- Freely moving electrophysiological recording with head/eye tracking.
- Molecular connectomics using barcoded rabies virus.

Postdoctoral fellow in the lab of Sonja Hofer, Biozentrum, Basel Switzerland, and Sainsbury Wellcome Centre - UCL, London

Functional organisation of higher-order thalamocortical circuits in the mouse visual system **February 2014–September 2020**

- Two-photon calcium imaging of axonal projections and acute silicon probe recording in awake mice.
- Posthoc immunostaining of cells functionally characterised by two-photon imaging in vivo.

Cerebellum Group, IBENS, Paris France

Function of interneurons in the molecular layer of the cerebellum **January 2009–Decembre 2013**

- Master 2 thesis (6 months), Ph.D. project (4 year) and postdoctoral position (3 months) under the supervision of Dr. Boris J. Barbour.
- Studying a local circuit using paired-recording, pharmacology, immunohistochemistry, modelling (rate model). Developed an analysis software in Python and Qt.

Billups Lab, Department of Pharmacology, University of Cambridge, Cambridge UK

Glutamine reuptake in neurons of the MNTB **January 2008–July 2008**

- Master 1 thesis under the supervision of Dr. Brian J. Billups.
- Studying glutamate-glutamine cycle using whole-cell patch-clamping and pharmacology. Learned to analyse and program with Igor Pro.

Laboratory of Neurobiology and Cellular Diversity, ESPCI, Paris France

Distribution of KATP channels in the rat barrel cortex **June and July 2007**

- Internship under the supervision of Dr. Bruno Cauli.
- Learned patch-clamp and single cell RT-mPCR.

EDUCATION **École Normale Supérieure**, Paris, France

Ph.D. in Neuroscience, ED3C, September 2009–2013

- Thesis Topic: *Function of molecular layer interneurons of the cerebellum*
- Adviser: Dr. Boris Barbour

Master in Neuroscience, Integrative Biology and physiology, 2007–2009

- With highest honours, ranking 2nd/50

Marine Biological Laboratory, Woods Hole, MA, USA

Summer school, Neurobiology, June and July 2012

- Intensive and comprehensive course of neurobiology

PUBLICATIONS

- A Chadwick, AG Khan, J Poort, **A Blot**, SB Hofer, TD Mrsic-Flogel, Maneesh Sahani. Learning shapes cortical dynamics to enhance integration of relevant sensory input. doi:<https://doi.org/10.1016/j.neuron.2022.10.001> *Neuron*, 2023.
- J Poort, KA Wilmes, **A Blot**, A Chadwick, M Sahani, C Clopath, Thomas D Mrsic-Flogel, Sonja B Hofer, Adil G Khan. Learning and attention increase visual response selectivity through distinct mechanisms. doi:[10.1016/j.neuron.2021.11.016](https://doi.org/10.1016/j.neuron.2021.11.016) *Neuron*, 2022.
- P Rupprecht, S Carta, A Hoffmann, M Echizen, **A Blot**, Alex C Kwan, Yang Dan, Sonja B Hofer, Kazuo Kitamura, Fritjof Helmchen, Rainer W Friedrich. A database and deep learning toolbox for noise-optimized, generalized spike inference from calcium imaging. *Nature neuroscience*, 2021. doi:<https://doi.org/10.1038/s41593-021-00895-5>
- A Blot***, MM Roth*, I Gasler, M Javadzadeh, F Imhof, SB Hofer. Visual intracortical and transthalamic pathways carry distinct information to cortical areas. *Neuron*, 2021. doi:<https://doi.org/10.1016/j.neuron.2021.04.017>
- Chabrol F P, **Blot A**, Mrsic-Flogel T D. Cerebellar contribution to preparatory activity in motor neocortex, *Neuron*, 2019. doi:[10.1016/j.neuron.2019.05.022](https://doi.org/10.1016/j.neuron.2019.05.022)
- Bouvier G, Aljadeff J, Clopath C, Bimbard C, Ranft J, **Blot A**, Nadal J-P, Brunel N, Hakim V, Barbour B. Cerebellar learning using perturbations. *Elife*, 2018. doi:[10.7554/eLife.31599.001](https://doi.org/10.7554/eLife.31599.001)
- Khan A*, Poort J*, Chadwick A*, **Blot A***, Sahani M, Mrsic-Flogel T D, Hofer S B. Distinct learning-induced changes in stimulus selectivity and interactions of GABAergic interneuron classes in visual cortex. *Nature neuroscience*, 2018. doi:[10.1038/s41593-018-0143-z](https://doi.org/10.1038/s41593-018-0143-z)
- Blot A***, de Solages, C*, Ostojic, S, Szapiro, G, Hakim, V & Léna, C. Time-invariant feed-forward inhibition of Purkinje cells in the cerebellar cortex in vivo. *The Journal of Physiology*, 2016. doi:[10.1113/JP271518](https://doi.org/10.1113/JP271518)
- Blot A** & Barbour B. Ultra-rapid axon-axon ephaptic inhibition of cerebellar Purkinje cells by the pinceau. *Nature Neuroscience*, 2014. doi:[10.1038/nn.3624](https://doi.org/10.1038/nn.3624)
- Blot A** & Barbour B. Analysis of the study of the cerebellar pinceau by Korn and Axelrad. *bioRxiv*, 2013. doi:[10.1101/001123](https://doi.org/10.1101/001123)
- Blot A**, Billups D, Bjørkmo M, Quazi AZ, Uwechue NM, Chaudhry FA, Billups B. Functional expression of two system A glutamine transporter isoforms in rat auditory brainstem neurons. *Neuroscience*, 2009. doi:[10.1016/j.neuroscience.2009.09.015](https://doi.org/10.1016/j.neuroscience.2009.09.015)

AWARDS AND
FELLOWSHIPS

EMBO Long term fellowship

May 2014

Two years-long fellowship co-funded by the EMBO (ALT 74-2014) and the European Commission FP7 (Marie Curie Actions, EMBOCOFUND2012, GA-2012-600394)

Élève de l'École Normale Supérieure

2006–2010

Fellowship for four years (élève fonctionnaire stagiaire) awarded after entrance in the École Normale Supérieure upon competitive examination

TEACHING
EXPERIENCE

TENSS, The Pike Lake, Romania

2019–present

- Teaching assistant for a week per year at the [Transylvanian Experimental Neuroscience Summer School](#) where Ph.D. students learn in vivo electrophysiology and behaviour.

Marine Biological Association of the UK, Plymouth, UK

2011–2019

- Instructor for [Microelectrode Techniques for Cell Physiology](#) on a whole-cell patch-clamping experiment during one week of this two-week workshop where Ph.D. students learn ex vivo electrophysiology.

University of Basel, Basel, CH

2014–2017

- Teaching assistant in “[Blockkurs: Cell Biology and Neurobiology](#)”, responsible for 2 days of practicals.

Ecole Normale Supérieure, Paris, France

2010–2013

- Co-responsible of *visual display of quantitative data* September 2013
 - 3 one-hour lectures, part of the *Micromodules : Insights in Life Sciences by SPIBens* program for 3rd year licence and 1st year masters students
- Instructor for *Interdisciplinary workshop: Mathematics and Programming for Biologists* September 2011 and 2012
 - Responsible for 1 week of lectures and practicals where 2nd year masters students are introduced to programming in Python and refresh their mathematics (matrix, eigenvalues/vectors, linear PDEs, probability and statistics ...)
- Instructor for *Biology in silico* November 2011 and 2012
 - Responsible for 16 hours of courses where 1st year masters students are introduced to programming in Python
- Co-organizer of *Professional Insertion* 2011–2013
 - Organiser of a series of 20 1-hour lectures given by Ph.D. students and postdocs of the IBENS where undergraduates (3rd year of licence) discover life after their degree.
- Instructor in *Practicals of genetics* Autumn 2010 (3 sessions)
 - Instructor in a practical course of genetics for junior undergraduates (3rd year licence) during three sessions of one week.

OTHER SKILLS

Languages:

- French (native language)
- English (fluent)

Computer skills:

- Extensive programming experience in Python and Matlab
- Familiar with Bonsai-rx, Qt, ImageJ, Igor
- Working knowledge of Neuron and R
- Other software: various acquisition and analysis software (Kilosort, PClamp, PatchMaster, WinWCP, ACQ4, microManager ...)

REFERENCES

Dr. Petr Znamenskiy (e-mail: petr.znamenskiy@crick.ac.uk;

- Group leader, Specification and Function of Neural Circuits Laboratory
The Francis Crick Institute
◇ 1 Midland Road, London NW1 1AT, UK

Dr. Boris Barbour (e-mail: boris.barbour@ens.fr; phone: +33-144-323-736)

- Chairman of Neuroscience section, group leader, Cerebellum group
Institut de Biologie de l'Ecole Normale Supérieure
◇ 46 rue d'Ulm, 75005 Paris, France

Dr. Sonja Hofer (e-mail: s.hofer@ucl.ac.uk; phone: +44 (0) 20 3108 8195)

- Group leader, [Sainsbury Wellcome centre - UCL - London](#)
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